

RK4 & Simpson's Rule — Quick Reference

A compact study sheet connecting Euler's approximation, Runge–Kutta 4th order (RK4), and Simpson's numerical integration rule.

1. Why does the slope change?

For an ODE, the derivative is generally a function of the current time and state:

$$dx/dt = f(t, x)$$

As x changes, $f(t,x)$ can change. Therefore the slope can change even in a perfectly controlled environment. Environmental disturbances are only one possible additional cause.

Coffee: $de/dt = -ke$. When the temperature error e is large, the cooling rate is large. As e approaches zero, the cooling rate approaches zero.

2. Euler's method

Euler evaluates the slope once, at the beginning of the timestep, and assumes that slope is approximately valid throughout the step.

$$x_{n+1} = x_n + dt \cdot f(t_n, x_n)$$

Mental model: Look at the current slope → assume it stays constant briefly → move → recalculate.

3. RK4: the central idea

RK4 samples the slope four times within one timestep. Later slope estimates use earlier estimates to predict where the state will be.

Stage	Formula	Meaning
k1	$f(t_n, x_n)$	Slope at start
k2	$f(t_n + dt/2, x_n + (dt/2)k1)$	Midpoint using k1
k3	$f(t_n + dt/2, x_n + (dt/2)k2)$	Improved midpoint
k4	$f(t_n + dt, x_n + dt \cdot k3)$	Slope at end

$$x_{n+1} = x_n + (dt/6) \cdot (k1 + 2k2 + 2k3 + k4)$$

Memory aid: Start → midpoint → midpoint again → end → weighted combination.

4. Why four stages?

k1: What is the slope now?

k2: If I follow k1 halfway, what is the slope there?

k3: Using the better midpoint estimate, what is the slope there?

k4: Using k3, what is the slope at the end?

Final: Combine them to estimate the average slope over the interval.

5. Simpson's rule

Simpson's rule estimates the area under a curve (a definite integral) using values at the start, midpoint, and end. It gives greater weight to the interior point.

$$\int[a \text{ to } b] f(x) dx \approx (b-a)/6 \cdot [f(a) + 4f((a+b)/2) + f(b)]$$

Weighting: 1 : 4 : 1. For composite Simpson's rule the pattern becomes 1, 4, 2, 4, 2, ..., 4, 1.

6. Simpson's rule and RK4

RK4 uses the related weighting pattern 1 : 2 : 2 : 1. The resemblance is useful intuition, but RK4 is **not simply Simpson's rule applied to f**. Its k2, k3, and k4 values are evaluated at predicted states.

7. Small comparison

For $dx/dt = x$, starting from $x = 1$ with $dt = 0.1$:

Method	After one step	Comment
Exact	1.1051709	Ground truth: $e^{0.1}$
Euler	1.1000000	Uses only the starting slope
RK4	1.1051708	Very close to exact

8. Euler vs RK4

Feature	Euler	RK4
Slope evaluations / step	1	4
Looks inside step?	No	Yes
Global error	$O(dt)$	$O(dt^4)$
Cost / step	Low	$\approx 4\times$ function evaluations
Main idea	Starting slope	Several predicted slopes

9. What does 4th order mean?

For smooth problems, Euler's global error scales roughly as dt , while RK4's global error scales roughly as dt^4 .

Halve $dt \rightarrow$ Euler error $\approx 1/2$ as large.

Halve $dt \rightarrow$ RK4 error $\approx 1/16$ as large.

These are asymptotic accuracy statements: they assume sufficiently small dt and sufficiently smooth dynamics.

10. Final memory map

Euler: One slope $\rightarrow$ one straight-line prediction.

RK4: Four intelligently predicted slopes $\rightarrow$ weighted average slope $\rightarrow$ one update.

Simpson: Start/middle/end values $\rightarrow$ weighted estimate of area under a curve.

Key idea: The slope changes because f depends on the changing state (and possibly time), not necessarily because the environment is irregular.